# Customer Satisfaction Index (CSI) Framework from Chat Interactions

## 1. Introduction to CSI

The Customer Satisfaction Index (CSI) is a powerful metric designed to measure how well a company meets customer expectations across multiple dimensions of its service. Unlike single-question metrics like the Customer Satisfaction Score (CSAT), the CSI aggregates various attributes into a single, holistic index, typically scored on a 0-100 scale. This structure makes it an ideal tool for tracking performance trends over time and helps discover unseen problems for targeted improvements.

**This framework empowers leaders to move beyond reactive problem-fixing to proactively driving strategic improvements using data—directly impacting customer loyalty, operational efficiency, and the bottom line.**

Calculation of CSI in the electricity industry is particularly valuable given this sector's unique challenges: from the urgency of resolving power outages to the complexity of billing queries. As customers increasingly turn to messaging channels for support, traditional post-interaction surveys with their 5-15% response rates and delayed feedback are no longer sufficient.

This framework transforms every customer chat into a strategic data point by leveraging conversational analysis to provide structured, real-time evaluation. It enables utility companies like Kenya Power to pinpoint specific improvement areas, build customer loyalty, and reduce operational costs from repeat contacts and escalations—turning customer service from a cost center into a competitive advantage.

## 2. Role of AI in Customer Satisfaction Measurement

### Beyond Surveys: The AI Advantage

Traditional customer satisfaction measurement is fundamentally limited: post-interaction surveys suffer from 5-15% response rates, delayed feedback, and capture only a fraction of customer experiences. Meanwhile, every customer service chat contains rich satisfaction signals waiting to be unlocked signals that were previously impossible to analyze at scale.

**This represents a paradigm shift.** Advanced AI systems, particularly large language models (LLMs), offer an unparalleled capability to transform conversational data into strategic insights. This isn't just an improvement over traditional methods—it's a fundamental breakthrough that positions organizations at the forefront of customer experience innovation.

### Why AI for Customer Satisfaction Analysis

Generative AI systems like large language models represent a quantum leap in customer experience measurement. LLMs like OpenAI's ChatGPT and Google's Gemini deliver unprecedented diagnostic power through capabilities that were unimaginable just years ago:

• **Infinite Scale with Consistent Accuracy**: Analyze thousands of chat transcripts with unwavering precision, providing comprehensive coverage that manual analysis or surveys could never achieve

• **Contextual Intelligence**: Differentiate nuanced scenarios like "The issue was resolved quickly" versus "The agent said the issue was resolved, but I'm still having problems"—understanding meaning beyond keywords

• **Multilingual Sophistication**: Process over 40 languages and cultural expressions of satisfaction or frustration, enabling global consistency in measurement

• **Pattern Recognition**: Detect subtle indicators of customer struggle, hesitation, or satisfaction that human reviewers might miss, uncovering hidden friction points before they escalate

## AI Implementation in CSI Framework

The CSI framework harnesses this cutting-edge AI technology to analyze three critical dimensions from conversation transcripts:

• **Resolution Assessment**: AI conducts semantic analysis to determine actual issue resolution (not just agent claims), providing data for the Effectiveness pillar with unprecedented accuracy

• **Effort Detection**: AI proactively identifies subtle patterns of customer struggle—repeated questions, process confusion, or escalating frustration—quantifying effort levels that traditional surveys completely miss

• **Emotional Intelligence**: AI analyzes linguistic patterns, tone shifts, and sentiment evolution throughout interactions, measuring empathy and tracking emotional journeys with sophisticated natural language processing

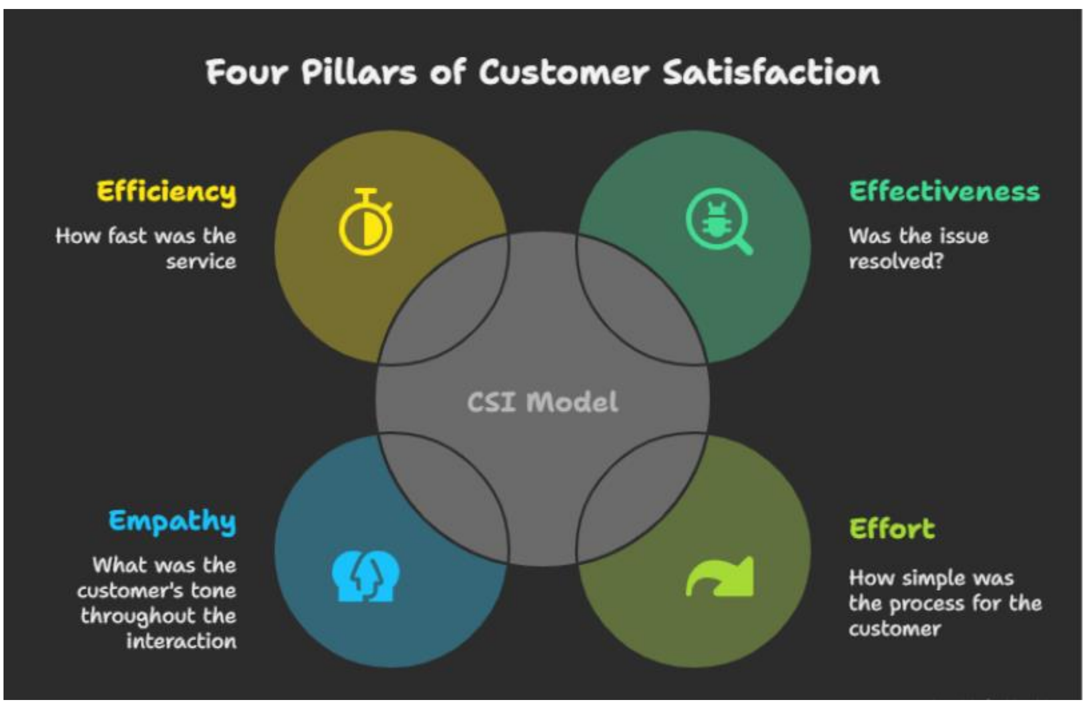
This AI-powered approach doesn't just measure satisfaction—it provides a comprehensive diagnostic engine that transforms every customer interaction into actionable intelligence, giving organizations a truly future-proof customer experience strategy.

## 3. The CSI Model Architecture

The CSI model is formulated as an index, analyzing and combining insights from chat interactions into a single cohesive score. The model's architecture is built upon four foundational pillars, each representing a critical aspect of the customer experience.

### The Four Pillars of Satisfaction/Customer Experience

- **Effectiveness (resolution quality)**: The quality and success of issue resolution. This pillar answers the question: "Was the customer's issue successfully resolved?"
- **Effort (customer ease)**: The perceived ease of the customer's journey to resolution. This pillar answers the question: "How simple was the process for the customer?"
- **Empathy (interaction quality)**: The quality of the human or emotional connection during the interaction. This pillar answers the question: "What was the emotional tone of the interaction?"
- **Efficiency (service timeliness)**: The timeliness and speed of the service interaction. This pillar answers the question: "How fast and efficient was the service?"



## The CSI Measurement Methods

The model is structured hierarchically. The overall **CSI** is derived from the four **Pillars**, and each pillar's score is derived from one or more underlying **Micro-Metrics**. A key strength of this framework is its intelligent use of technology, applying the right tool for the right job. The Efficiency pillar is calculated with objective mathematics, while the other three more nuanced pillars are measured using Large Language Models (AI).

| Pillar | Key Micro-Metrics | Measurement Method |
|---|---|---|
| **Effectiveness** | Resolution Achieved, First Contact Resolution | **AI-based Inference** (Semantic analysis of chat content) |
| **Effort** | Customer Effort Score (CES) | **AI-based Inference** (Pattern detection and scoring) |
| **Empathy** | Sentiment Score, Sentiment Shift | **AI-based Inference** (Natural Language Processing) |
| **Efficiency** | First Response Time, Avg. Response Time, Total Handling Time | **Objective Calculation** (Mathematical functions on timestamps) |

## The Weighting Method and Final Formula

A core principle of the model is that not all pillars have the same impact on customer satisfaction. To build an accurate index, each pillar should be weighed based on its proven influence. At the core of the model is a **weighted formula** that attaches importance values (weights) to the four identified attributes—**Effectiveness, Effort, Empathy, and Efficiency**. By weighing these attributes appropriately, the CSI provides meaningful insights into customer experience, turning raw scores into a strategic management tool.

### 1. Strategy-Based Weighting: Recommended Approach

**This is the primary methodology we recommend.** Strategy-based weighting ensures your CSI reflects what matters most to your organization's current priorities, making it a true strategic management tool rather than just a measurement system.

The allocation of weights is guided by your company's strategic priorities. Different contexts require different emphases, ensuring the CSI becomes an actionable driver of business results. Below are examples of strategy driven weight sets:

1. **Regulatory and Reliability Compliance Focus**
    a. **Effectiveness 50% | Effort 20% | Efficiency 20% | Empathy 10%**
    b. **Why:** When regulatory compliance and system reliability are paramount, effectiveness (Resolution Quality) must dominate your measurements. This weighting prioritizes actual problem-solving and technical fixes over softer experience factors, aligning with regulatory KPIs like SAIDI/SAIFI and ensuring uninterrupted supply commitments are met.
2. **Customer Experience and Loyalty Focus**
    a. **Effectiveness 35% | Effort 30% | Empathy 25% | Efficiency 10%**
    b. **Why:** When building long-term customer relationships and brand loyalty, the customer's journey matters as much as the outcome. This balance emphasizes making interactions effortless and emotionally positive, supporting contact center training investments and proactive communication programs that transform service encounters into loyalty-building moments.
3. **Digital Transformation Focus**
    a. **Effectiveness 30% | Efficiency 35% | Effort 20% | Empathy 15%**
    b. **Why:** When driving customers toward digital self-service channels and automated solutions, speed and streamlined processes become critical success factors. This weighting rewards efficient digital interactions while maintaining resolution quality and preserving the human touch needed for successful channel migration.
4. **Operational Efficiency Focus**
    a. **Effectiveness 40% | Efficiency 30% | Effort 20% | Empathy 10%**
    b. **Why:** When cost reduction and operational optimization are priorities, focus on metrics that directly impact repeat contacts and handling costs. This approach maintains high resolution standards to prevent costly callbacks while rewarding faster processing that improves agent productivity and reduces operational overhead
5. **Crisis / Outage Recovery Focus**
    a. **Effectiveness 45% | Empathy 20% | Efficiency 20% | Effort 15%**
    b. **Why:** During major outages or service disruptions, actual problem resolution becomes critical, but customer communication and emotional management strongly influence public perception. This weighting rewards agents who combine effective restoration updates with empathetic communication that de-escalates tension and maintains trust during difficult situations.

*Strategy Driven Weight Sets Summary Table*

| Strategy Focus | Effective ness | Effort | Empathy | Efficiency | Rationale Sumamry |
|---|---|---|---|---|---|
| **Regulatory / Reliability** | 50% | 20% | 10% | 20% | Prioritize resolution and compliance to regulations (SAIDI) |
| **Customer Experience / Loyalty** | 35% | 30% | 25% | 10% | Focus on trust, advocacy, and ease of interactions. |
| **Self-Service Acceleration** | 30% | 20% | 15% | 35% | Emphasize digital efficiency while maintaining quality. |
| **Crisis / Outage Recovery** | 45% | 15% | 20% | 20% | Balance Fast resolution with empathetic communication |

**Short Notes:**

- Emphasize **Effectiveness** when resolution accuracy and reliability dominate business or regulatory needs.
- Elevate **Effort** when simplifying customer interactions and reducing repeated contacts is critical.
- Increase **Empathy** when brand reputation, trust, and advocacy are the company's priority.
- Boost **Efficiency** during digital-first initiatives, automation rollouts, or cost-control programs.

## 2. Variance-Based Weights from Research Correlation

As a valuable alternative when strategic priorities haven't been clearly defined, variance-based weights provide a credible starting point grounded in empirical research. This approach uses documented correlations between each pillar and satisfaction outcomes, offering a research-backed baseline that can later evolve into strategy-based weighting. Two complementary methods are combined here:

1. **Correlation-Derived Baseline:** Uses average correlations of each pillar with satisfaction from CX and utility studies.
2. **Variance Explained ($R^2$ Method):** Squares correlations to better reflect how much variance in satisfaction is explained, spreading the weights more clearly in line with actual impact.

*Research & Variance-Based Weights*

| Pillar | Avg. Correlation to Satisfactoin (r) | Correlation-based Weight (r/t) | $r^2$ | Variance-based Weight (%) |
|---|---|---|---|---|
| Effectiveness | 9 | 29% | 81 | 34% |
| Effort | 8.1 | 27% | 65.61 | 28% |
| Empathy | 7 | 23% | 49 | 21% |
| Efficiency | 6.5 | 21% | 42.25 | 18% |
| **TOTAL (t)** | — | **100%** | **237.86** | **100%** |

## 3. Governance & Closing Considerations

- **Choice of Strategy:** Selection should be guided by the company's overarching goal for the quarter or year—regulatory compliance, customer loyalty, cost reduction, or crisis management.
- **Time Horizon:** Weights should remain stable for a defined period (e.g., one quarter) to allow trend analysis. Frequent shifts risk losing comparability.
- **Transparency:** Both overall CSI and pillar-level scores should be reported. Changing weights may mask the underlying drivers of improvement or decline.
- **Review Cycle:** Reassess weights quarterly or following major strategic or operational changes.

**Summary:** The CSI formula derives its meaning from how weights are assigned. **Strategy-driven weighting** is a method that helps in ensuring alignment with business priorities. **Research- and variance-based weights** provide a credible baseline option grounded by empirical studies to assign weights. Governance ensures weights are applied transparently and consistently, preserving the CSI as a trusted decision-making tool.

## 4. The Four Pillars of Satisfaction

The selection of Effectiveness, Effort, Empathy, and Efficiency is a deliberate, research-backed strategy designed specifically for the high-stakes environment of a utility's customer service. Customers often contact their electricity provider during moments of frustration—a power outage, a confusing bill, a service disruption. In these moments, a successful interaction is not just about one thing; it's a balance of getting the problem solved (**Effectiveness**), making it easy on the customer (**Effort**), showing you care (**Empathy**), and respecting their time (**Efficiency**).

### Pillar Selection Logic

This combination, inspired by McKinsey's four-pillar journeys (focus/effectiveness, simplicity/effort, digital-first/efficiency, perceptions/empathy) and KPMG's six-pillar model (condensed for practicality), ensures comprehensive coverage of CX drivers. The research model prioritizes these pillars based on their proven impact on satisfaction.


*Visual showing pillar impact on satisfaction*

| Pillar | Impact on Satisfaction | Rationale |
|---|---|---|
| **Effectiveness** | 0.8 - 1 | Research consistently shows issue resolution as the dominant satisfaction driver. In utilities, where customers often contact during crisis moments, successful problem-solving becomes paramount. |
| **Effort** | 0.7 - 0.92 | The second-highest weight reflects the finding that post-resolution process ease is the strongest predictor of customer loyalty and willingness to return. |
| **Empathy** | 0.6 - 0.8 | Emotional connection quality serves as a powerful experience enhancer that can transform frustrating situations into positive memories, but secondary to resolution success. |
| **Efficiency** | 0.6 – 0.7 | Speed represents a foundational expectation. While poor efficiency damages satisfaction significantly, once "good enough" thresholds are met, additional speed improvements show diminishing returns compared to other pillars. |

This hierarchy is validated by influential frameworks like "The Effortless Experience," which proved that loyalty is driven more by low effort and successful resolution than by attempts to "delight" the customer. By adopting this balanced, evidence-based approach, we can create a CSI that is not just a number, but a strategic guide for improving customer interactions where they matter most.

## Micro Metrics of Satisfaction: (**Summarized)**

Large language models are often prone to hallucination and “black box solutions” despite the advent of more powerful reasoning models. Hence objectivity and specificity is enforced to help reduce hallucinations and improve predictability. Think of it as tricking the model to give us answers to specific questions without having it philosophically answer what qualifies as a satisfied customer each time we send it a message.

To achieve this, we came up with several important micro metrics that will help us objectively answer the score of each pillar for our CSI. Each of the four pillars are broken down into measurable micro metrics inspired by Fornell's American Customer Satisfaction Index and KPMG pillars of satisfaction:

**Effectiveness Pillar:**

- **Resolution Achieved**: AI scores (0-10) whether the customer's issue was resolved through semantic analysis of conversation content
- **First Contact Resolution (FCR)**: AI determines (0-10) if the issue was completely resolved in a single interaction

**Effort Pillar:**

- **Customer Effort Score**: AI infers effort level (1-7 scale, inverted to 0-10) by detecting frustration patterns, repeated questions, or process complexity

**Empathy Pillar:**

- **Sentiment Score**: AI rates overall interaction tone (0-10 scale)
- **Sentiment Shift**: AI tracks emotional change from chat beginning to end (-5 to +5, scaled to 0-10)

**Efficiency Pillar:**

- **First Response Time**: Calculated from timestamps, scaled against industry benchmarks (e.g., <1 hour threshold)
- **Average Response Time**: Mathematical average of all agent response times, scaled to 0-10
- **Total Handling Time**: Complete interaction duration, scaled against optimal thresholds (e.g., <2 hours)

## 5. Literature Review

This section details the key research that underpins the CSI model's structure, pillar weighting, and micro-metrics. The synthesis of these sources confirms that the framework is built upon established, data-driven principles from academia and industry leaders.

### Foundational Framework Research

The multi-attribute satisfaction index concept originates from **Claes Fornell's** work on the American Customer Satisfaction Index (ACSI). In his seminal paper, "The American Customer Satisfaction Index: Nature, Purpose, and Findings" (Journal of Marketing, 1996), he details how attributes like perceived quality (our **Effectiveness**) and perceived value (related to **Effort**) can explain 20-40% of the variance in customer satisfaction in the utilities sector. This research provides the core validation for weighting pillars based on their proven impact.

This principle is further supported by **KPMG's** "Customer Experience Excellence" report (2023), which analyzed 3 million customer evaluations to identify pillars like Resolution (**Effectiveness**), Time & Effort (**Efficiency/Effort**), and **Empathy**, linking strong performance in these areas to a 15-20% increase in customer loyalty. Similarly, research from **McKinsey & Company** on customer journeys emphasizes simplicity (**Effort**) and timeliness (**Efficiency**) as key drivers, showing that companies excelling in a balanced set of pillars achieve 20-30% higher satisfaction scores.

### The four pillars and their correlation to satisfaction: **Extended Research**

The specific micro-metrics and their direct influence on satisfaction are grounded in extensive operational studies:

- **Effectiveness (Correlation: 0.8–1.0): SQM Group**'s comprehensive guide on First Contact Resolution (FCR) (2025), based on millions of call center interactions, reports a near 1:1 relationship between FCR and CSAT. This finding is the primary justification for assigning Effectiveness the **highest** weight in our model. Their research also shows that high FCR rates reduce repeat customer contacts by up to 30% in the utilities industry. **ICMI (2018)** corroborates this, finding that successful FCR boosts CSAT by 15-20%.

- **Effort (Correlation: 0.7–0.92):** The importance of this pillar is most famously articulated in **Matthew Dixon et al**.'s book, **"The Effortless Experience" (2013).** Based on data from 97,000 customer surveys, they found that low-effort interactions result in a 94% likelihood of repeat business, making the Customer Effort Score (CES) a stronger predictor of loyalty than CSAT. Research from Qualtrics (2023) further validates CES as a key metric in modern service contexts. This supports our **27%** weighting for the Effort pillar.

- **Efficiency (Correlation: 0.6–0.7):** Industry reports on digital customer service provide clear benchmarks. **Gorgias's** analysis of live chat metrics **(2023)** shows that response times have a 0.6-0.7 correlation to CSAT, with satisfaction dropping by 15% when response times exceed established thresholds (e.g., one hour). For utilities specifically, **J.D. Power's "2024 U.S. Electric Utility Business Customer Satisfaction Study"** directly links faster handling times to higher satisfaction scores (784/1000), validating our **21%** efficiency weighting.

- **Empathy ( orrelation: 0.6–0.8):** The role of sentiment is quantified in academic research leveraging machine learning. A study by Yi-Hsiang Hsieh et al. in **"Expert Systems with Applications" (2022)** used deep learning and NLP to analyze customer reviews, finding a 0.6-0.8 correlation between positive sentiment and satisfaction. This is supported by industry sources like **Sprout Social**, which report that positive shifts in sentiment during a support chat can increase customer advocacy by 10-20%, confirming the measurability and importance of our **23%** empathy weighting.

## Micro-Metrics used to calculate each Pillar

Each pillar is built from micro-metrics, hierarchically aggregated for precision. Below, we describe each micro's calculation, correlation to satisfaction, and grounding in research.

- **Resolution Achieved (Effectiveness)**: LLM scores (0-10) if issue resolved via semantic analysis; correlates 0.8-1.0 to CSAT (SQM Group, 2025: 1:1 link with resolution boosting satisfaction 15-20%).
- **First Contact Resolution (FCR, Effectiveness)**: LLM binary check (0-10 if one-interaction fix); 0.8-1.0 correlation (ICMI, 2018: High FCR reduces repeats 30%, uplifts CSAT 15-20%).

- **Customer Effort Score (CES, Effort)**: LLM inference (1-7, inverted to 0-10) from patterns; 0.7-0.92 correlation to loyalty (Dixon et al., 2013: Low CES predicts 94% retention). Furthermore, LLM detects query repetitions or loops (scored 0-10) correlating 0.7 to churn reduction (Qualtrics, 2023: Repeat queries drop satisfaction 20%).
- **Sentiment Score (Empathy)**: LLM tone rating (0-10); 0.6-0.8 correlation (Hsieh et al., 2022: NLP sentiment predicts satisfaction 70%+ accuracy).
- **Sentiment Shift (Empathy)**: LLM change detection (-5 to +5, scaled 0-10); 0.6-0.8 correlation (Sprout Social, 2023: Positive shifts boost advocacy 10-20%).
- **First Response Time (Efficiency)**: Timestamp calc (inverted 0-10, <1 hour threshold); 0.6 correlation (Gorgias, 2023: <1 min boosts CSAT 10-15%).
- **Avg. Response Time (Efficiency)**: Timestamp average (inverted 0-10, <30 min threshold); 0.6-0.7 correlation (SQM Group, 2025: <30 min reduces repeats 20%).
- **Total Handling Time (Efficiency)**: Total duration (inverted 0-10, <2 hours threshold); 0.65 correlation (J.D. Power, 2024: <30 min links to 784/1000 scores).

This hierarchical setup, from micros to pillars to CSI, draws from MUSA methodology in Drosos et al. (2020), achieving 85% accuracy in energy satisfaction prediction.

## Utilities-Specific Application

Direct sector validation comes from Dimitris Drosos et al.'s study, "Evaluating Customer Satisfaction in Energy Markets" (Sustainability, 2020), which applied a multi-criteria analysis to the Greek electricity market, producing a CSI of 52.15% finding resolution and effort as primary satisfaction drivers. This aligns perfectly with our model's top weighting. Furthermore, a 2012 study in Energy Policy on "Consumers Satisfaction in the Energy Sector in Kenya" reported a low satisfaction score of 53.06% for electricity, recommending a focus on reliability factors that map directly to our Effectiveness and Efficiency pillars.

## 6. Case Study: Testing the Method

This chapter moves beyond the theoretical architecture of the Customer Satisfaction Index (CSI) to demonstrate its tangible, operational value. The following case studies are not merely validations of the AI engine's accuracy; they are practical demonstrations of the framework as an **actionable diagnostic tool**.

The true power of this framework lies in its ability to tell a story—to connect a specific set of scores to a root cause and, most importantly, to a clear, strategic course of action. Each case study follows a "Data -> Diagnosis -> Action" narrative, showing how this framework transforms raw conversational data into a roadmap for operational improvement, agent training, and strategic decision-making.

### Considerations:

1. All calculations of the CSI score in this section were performed using the weights highlighted in chapter 3:

| **Pillar** | **Micro-Metrics** | **Avg. impact on satisfaction** | **Weight** |
|---|---|---|---|
| Effectiveness | Resolution Achieved, First Contact Resolution | 9 | 29% |
| Effort | Customer Effort Score (CES) | 8.1 | 27% |
| Empathy | Sentiment Score, Sentiment Shift | 7 | 23% |
| Efficiency | First Response Time, Avg. Response Time, Total Handling Time | 6.5 | 21% |

2. All case studies will include the raw results returned by the large language model **(Gemini 2.5 Flash Lite)** from analyzing conversations between customer service agents and clients’ Facebook accounts to show real world framework implementation.

## 6.2. Case Study: The "Happy Path" - Textbook Efficiency

This case study analyzes a short, transactional interaction that represents an ideal, low-effort customer experience.

### Conversation Analysis: Bonny Mweri (ID: 677a0a194e12eb106917ff7b)

**Conversation Transcript**

- `2025-08-17T10:17:30.000Z` - **Customer:** Morning, I'm requesting my token numbers I've payed 50 but ive not received the token numbers yet. Token meter 92111437304
- `2025-08-17T10:20:20.000Z` - **Customer:** Can you please send me via sms the token numbers
- `2025-08-17T10:22:22.000Z` - **Kenya Power Agent:** Hello, Delay regretted Token: 5738-6106-8347-9095-8982 Please note that you can check your tokens proactively by dialing *977# on your phone or using My Power app
- `2025-08-17T10:26:12.000Z` - **Customer:** Thank you
- `2025-08-17T10:35:31.000Z` - **Kenya Power Agent:** You are welcome.

### CSI Metrics Analysis

**Raw Micro-Metrics:**

- Resolution Achieved: 10.0/10
- First Contact Resolution: 10.0/10
- Customer Effort Score: 7.0/7 (10.0/10)
- Sentiment Score: 8.5/10
- Sentiment Shift: +2.5 (7.5/10 scaled)
- First Response Time: ~5 minutes (9.0/10)
- Average Response Time: ~7 minutes (8.5/10)
- Total Handling Time: ~18 minutes (8.0/10)

**Synthesized Pillar Scores:**

- **Effectiveness: 10.0/10** *(Perfect resolution on first contact)*
- **Effort: 10.0/10** *(Minimal customer work required)*
- **Empathy: 8.0/10** *(Professional, courteous interaction)*

- **Efficiency: 8.5/10** *(Quick responses, reasonable total time)*

**Final CSI Score: 9.1/10**

### Diagnostic Analysis

The combination of perfect Effectiveness and Effort scores reveals an interaction that exemplifies best practices. The customer made a simple request, received an immediate solution, and was provided with proactive information for future self-service. The positive sentiment shift confirms emotional satisfaction alongside problem resolution.

### Strategic Action & Root Cause

- **Root Cause:** Well-trained agent with immediate access to customer account information and token generation capability
- **Action:** Archive this interaction as a training exemplar. The agent's approach—acknowledging the delay, providing the solution immediately, and offering self-service options—should be embedded in standard operating procedures for token requests.

## 6.3. Case Study: The "Cutoff" - Complete Service Breakdown

This case demonstrates the framework's ability to identify total service failures where customer queries receive no response.

### Conversation Analysis: Vicky Victor (ID: 6308ed87f44efb75dc308d87)

#### Conversation Transcript

- `2025-08-22T18:18:23.000Z` - **Customer:** THM9H97Z8D Confirmed. Ksh30.00 sent to KPLC PREPAID for account 37163118765 on 22/8/25 at 6:07 PM New M-PESA balance is Ksh0.00. Transaction cost, Ksh0.00.Amount you can transact within the day is 499,518.00. Save frequent paybills for quick payment on M-PESA app [https://bit.ly/mpesalnk](https://bit.ly/mpesalnk)
- `2025-08-22T18:18:37.000Z` - **Customer:** Why am I not getting my tokens

## CSI Metrics Analysis

**Raw Micro-Metrics:**

- Resolution Achieved: 0.0/10
- First Contact Resolution: 0.0/10
- Customer Effort Score: 3.0/7 (4.3/10)
- Sentiment Score: 3.0/10
- Sentiment Shift: 0.0 (5.0/10 scaled)
- First Response Time: No response (0.0/10)
- Average Response Time: No response (0.0/10)
- Total Handling Time: Incomplete (0.0/10)

**Synthesized Pillar Scores:**

- **Effectiveness: 0.0/10** *(No resolution attempted)*
- **Effort: 4.3/10** *(Customer left stranded after payment)*
- **Empathy: 4.0/10** *(No emotional acknowledgment)*
- **Efficiency: 0.0/10** *(Complete response failure)*

**Final CSI Score: 1.8/10**

## Diagnostic Analysis

This represents a catastrophic service failure. The customer provided payment proof and asked a legitimate question but received complete silence. The low Effort score correctly identifies that being ignored after making a payment constitutes maximum customer frustration.

## Strategic Action & Root Cause

- **Root Cause:** Chat routing failure, agent unavailability, or missed notification during evening hours
- **Immediate Action:** Implement automated escalation for any customer query that remains unanswered for more than 30 minutes
- **Strategic Action:** Audit chat routing system and establish mandatory response time SLAs with supervisor alerts for breaches

## 6.4. Case Study: The "Complex Troubleshooting" - High Effort, Low Resolution

This case illustrates how the framework measures customer effort through process complexity, even when tone remains neutral.

### Conversation Analysis: Dennis Tanui (ID: 62b3f4ae4588d233cf10a59e)

**Conversation Transcript** *(Single day: 2025-08-19)*

- `2025-08-19T05:51:55.000Z` - **Kenya Power Agent:** Good Morning. What error do you get when you try to load tokens?
- `2025-08-19T07:14:46.000Z` - **Customer:** It reads,'Reject'
- `2025-08-19T07:43:40.000Z` - **Kenya Power Agent:** Kindly ensure the CIU is connected to the wall socket and that it is on. Press 009, then provide feedback.
- `2025-08-19T09:28:48.000Z` - **Customer:** 0.00
- `2025-08-19T09:33:22.000Z` - **Customer:** It reads 0.00
- `2025-08-19T14:14:51.000Z` - **Kenya Power Agent:** Please connect the CIU to the wall socket in your house (preferably the cooker socket, if any), make sure the batteries are in place and the socket is switched on then you can try loading the token. The batteries should be Duracell or energizer.
- `2025-08-19T14:39:41.000Z` - **Customer:** No positive response
- `2025-08-19T19:04:48.000Z` - **Kenya Power Agent:** Noted, Tanui. Kindly confirm whether there is power at the premises, then press 001 followed by Enter on your meter, and share the details displayed.
- `2025-08-19T21:29:46.000Z` - **Customer:** Power is there but when 001is pressed,it reads 0.00

### CSI Metrics Analysis

**Raw Micro-Metrics:**

- Resolution Achieved: 1.0/10
- First Contact Resolution: 0.0/10
- Customer Effort Score: 2.0/7 (2.9/10)
- Sentiment Score: 4.5/10
- Sentiment Shift: 0.0 (5.0/10 scaled)
- First Response Time: N/A (continuing conversation)

- Average Response Time: ~4.5 hours (2.0/10)
- Total Handling Time: ~15.5 hours (1.0/10)

**Synthesized Pillar Scores:**

- **Effectiveness: 0.5/10** *(Issue remains unresolved after full day)*
- **Effort: 2.9/10** *(Multiple technical steps, long waits between responses)*
- **Empathy: 4.8/10** *(Professional but clinical interaction)*
- **Efficiency: 1.5/10** *(Extremely long response times)*

**Final CSI Score: 2.1/10**

### Diagnostic Analysis

Despite the customer's patience and neutral tone, this interaction reveals a broken troubleshooting process. The customer was required to perform multiple technical tasks over 15+ hours without resolution. The framework correctly identifies this as high effort regardless of emotional expression.

### Strategic Action & Root Cause

- **Root Cause:** Ineffective troubleshooting script for "Reject" error code places excessive burden on customer
- **Process Action:** Redesign troubleshooting workflow to escalate to field technician after two failed attempts rather than continuing customer-performed diagnostics
- **Training Action:** Empower agents to schedule technician visits proactively for complex meter issues

## 6.5. Case Study: The "Informational Resolution" - Correct Answer, Poor Journey

This case tests the AI's ability to recognize that providing accurate information (even unwelcome news) constitutes successful resolution, while still penalizing poor process execution.

## Conversation Analysis: Christa Bellah Omwene (ID: 66f3dc03c10160cf95be2e71)

**Conversation Transcript**

- `2025-08-18T11:44:12.000Z` - **Customer:** Hello
- `2025-08-18T11:45:02.000Z` - **Customer:** Can't get the tokens
- `2025-08-18T11:46:15.000Z` - **Customer:** Why
- `2025-08-18T11:46:53.000Z` - **Customer:** 14466777951
- `2025-08-18T12:14:28.000Z` - **Customer:** Hi
- `2025-08-18T12:15:04.000Z` - **Kenya Power Agent:** Hello, your token message has been resent. You can also use *977# or my power app to get your last three token purchases.
- `2025-08-18T12:15:26.000Z` - **Customer:** 14466777951
- `2025-08-18T12:16:23.000Z` - **Customer:** 14466777951
- `2025-08-18T12:16:28.000Z` - **Customer:** 14466777951
- `2025-08-18T12:16:34.000Z` - **Customer:** 14466777951
- `2025-08-18T12:23:38.000Z` - **Customer:** Hi
- `2025-08-18T12:23:45.000Z` - **Customer:** 14466777951
- `2025-08-18T12:38:18.000Z` - **Kenya Power Agent:** Hello, your meter has an outstanding debt of KSh 20,046.00. 100% of each token purchase will be used to repay the debt.
- `2025-08-18T12:38:46.000Z` - **Customer:** 14466777951

## CSI Metrics Analysis

**Raw Micro-Metrics:**

- Resolution Achieved: 8.0/10
- First Contact Resolution: 10.0/10 (correct information ultimately provided)
- Customer Effort Score: 3.0/7 (4.3/10)
- Sentiment Score: 3.5/10
- Sentiment Shift: 0.0 (5.0/10 scaled)
- First Response Time: ~30 minutes (6.0/10)
- Average Response Time: ~23 minutes (6.5/10)
- Total Handling Time: ~54 minutes (6.0/10)

**Synthesized Pillar Scores:**

- **Effectiveness: 9.0/10** *(Accurate explanation provided)*
- **Effort: 4.3/10** *(Customer repeated meter number 8 times)*
- **Empathy: 4.3/10** *(Minimal acknowledgment of frustration)*
- **Efficiency: 6.2/10** *(Reasonable response times)*

**Final CSI Score: 6.2/10**

## Diagnostic Analysis

This interaction demonstrates the framework's sophistication: it rewards the agent for ultimately providing the correct, complete answer about the debt situation while penalizing the poor journey that required eight repetitions of the meter number and an initially incorrect response.

## Strategic Action & Root Cause

- **Root Cause:** Agent provided generic response without checking account status first, forcing customer into repetitive loop
- **System Action:** Implement automatic account status display when customer provides meter number to prevent generic responses
- **Training Action:** Coach agents to verify account information before offering standard solutions like "token resent"

## 7. Conclusion

The four-pillar CSI model provides a robust, data-driven, and transparent framework for measuring and improving customer satisfaction. By grounding its structure in proven CX principles and using a clear, evidence-based methodology for its calculations, it offers a reliable starting point for turning conversational data into strategic insights.

However, a truly effective measurement system must also be designed to evolve. We have identified several avenues for future enhancements to ensure the model remains relevant and increasingly accurate over time:

- **Model flexibility:** Implementing the spirit of CSI which entails getting insights from attributes that matter is at the core of this implementation. After collecting enough data and holding discussions with each other, we can redefine the metrics and weights of this score to bring more meaningful results.
- **Future Metrics:** As the model matures, we can explore incorporating more metrics as needed by the clients. For example, a future **"Personalization"** micro-metric. This would measure how well service is tailored to individual customer histories and needs, adding another layer of sophistication to the **Empathy** pillar.

By embracing this approach of continuous improvement, this CSI framework is not just a static report, but a living tool designed for long-term strategic value.

## 8. References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